

SMART LOCKER SYSTEM

Operation Manual

Biometric Access · WebAuthn · ESP32

Website: smart-locker-backend-o5ha.onrender.com

Operation Video: [Watch on Google Drive](#)

Table of Contents

1. System Overview	4
1.1 How It Works	4
1.2 Hardware Components	4
1.3 System Architecture	5
2. Getting Started	6
2.1 What You Need	6
2.2 Opening the Website	6
3. Registering (First-Time Setup)	7
3.1 Step-by-Step Registration.....	7
3.2 Supported Authentication Methods.....	7
4. Claiming a Locker.....	9
4.1 Returning Users	9
4.2 Step-by-Step — Claiming	9
4.3 What If No Lockers Are Available?.....	10
5. Using Your Locker.....	11
5.1 Unlocking (Opening) the Locker	11
5.2 Locking the Locker	11
5.3 Checking the LCD Screen	12
6. Returning (Releasing) Your Locker.....	13
6.1 Step-by-Step — Returning	13
7. Session Restore (Page Refresh)	14
7.1 How It Works	14
7.2 What To Do If Your Session Is Not Restored.....	14
8. Locker Status Lifecycle.....	15

8.1 Automatic Timeouts	15
9. Troubleshooting	16
9.1 Common Issues and Solutions	16
10. Quick Reference Card	18

1. System Overview

The Smart Locker System is a biometric-secured storage solution that replaces physical keys with your phone's built-in security (fingerprint, Face ID, PIN, or pattern). It uses WebAuthn - so your biometric data never leaves your device.

Key Benefit: You never need a physical key. If you lose access to a locker, there is no fine — simply re-authenticate from any device.

1.1 How It Works

- 1. Register:** Set up your biometric credential once on the website.
- 2. Claim:** Log in and claim a locker — the system assigns the next available one automatically.
- 3. Unlock / Lock:** Open and close your locker from the web interface at any time during your session.
- 4. Release:** When finished, press "I'm Done / Return Locker" to free the locker for the next user.

1.2 Hardware Components

The physical locker cabinet contains:

- ESP32 microcontroller — the brain of the locker
- 2-channel relay module — switches the solenoid locks on / off
- Solenoid lock (per locker bay) — bolt goes DOWN to open, UP to close
- 16 × 2 I²C LCD screen — shows locker status and assigned user

- Battery pack — powers the ESP32 and relay

1.3 System Architecture

The system consists of three layers that communicate in real time:

Browser → WebAuthn authentication + REST commands (claim, unlock, lock, release)

Backend Server → Node/Express + SQLite assigns lockers and sends MQTT commands

ESP32 Hardware → Receives MQTT commands and drives the relay / LCD accordingly

Privacy Notice: We do not store your fingerprint, Face ID, PIN, or biometric data of any kind. We only store a unique user ID so you can claim lockers on future visits without re-registering.

2. Getting Started

2.1 What You Need

- A smartphone or laptop with biometric capability (fingerprint sensor, Face ID, or a PIN/pattern)
- A supported browser: Chrome, Edge, Safari, or Firefox (latest version recommended)
- An internet connection

2.2 Opening the Website

Open your browser and navigate to:

Smart Locker Web App: <https://smart-locker-backend-o5ha.onrender.com>

You can also **bookmark this page** on your phone for quicker access on future visits.

Video Walkthrough: A full video demonstration of the operation process is available at:

<https://drive.google.com/file/d/1R2BMD9DogCVqviRVV4EwPOxApFnTIgfG/view?usp=sharing>

3. Registering (First-Time Setup)

Registration is a one-time process. Once registered, you can claim lockers on future visits without repeating this step.

3.1 Step-by-Step Registration

Step	Action
1	Open the Smart Locker website on your device.
2	Click or tap "Register" on the homepage.
3	Your browser will prompt you to verify your identity using your device's built-in security (fingerprint, Face ID, PIN, or pattern).
4	Complete the biometric or PIN prompt on your device.
5	Once verified, your account is created. You will see a confirmation on screen.

Important: You must register on each device you intend to use (e.g. your phone AND your laptop are treated as separate registrations). However, once registered on a device, you do not need to register again on that device.

3.2 Supported Authentication Methods

The system accepts any method your device supports via WebAuthn:

- Fingerprint sensor
- Face ID / Face recognition
- Device PIN or password

- Pattern unlock (Android)

Your biometric data never leaves your device and is not transmitted to or stored on our servers.

4. Claiming a Locker

After registering (or if you are a returning user), follow these steps to get a locker assigned to you.

4.1 Returning Users

If you have registered before, you do not need to re-register. Simply go to the website and proceed to Step 3 below — you will be prompted to verify your identity and then go straight to claiming.

4.2 Step-by-Step — Claiming

Step	Action
1	Open the Smart Locker website.
2	Verify your identity with your biometric / PIN when prompted (this confirms who you are).
3	Press "Claim Locker".
4	The system automatically assigns you the next available locker (e.g. "You have been assigned Locker 1").
5	The LCD screen on the locker cabinet updates to show your assignment.

Time Limit: You have 5 minutes after claiming before you physically unlock the locker. If you do not unlock within 5 minutes, the locker is automatically released back to the available pool. You will need to claim again.

4.3 What If No Lockers Are Available?

The system will notify you if all lockers are currently occupied. Wait until another user releases their locker and try again. The system automatically frees abandoned lockers after a timeout, so availability can change at any moment.

5. Using Your Locker

5.1 Unlocking (Opening) the Locker

Step	Action
1	On the web interface, press "Unlock".
2	You will hear a click — the solenoid bolt moves DOWN, releasing the door.
3	The LCD screen updates to show "Locker [N] Unlocked".
4	Open the locker door and place your belongings inside.
5	Close the door when done.

Note: The bolt going DOWN means the locker is OPEN. Bolt UP means LOCKED. This is visible on the physical locker.

5.2 Locking the Locker

Step	Action
1	Press "Lock" on the web interface.
2	The solenoid bolt moves UP — the door is now secured.
3	The LCD screen updates to show "Locker [N] Locked".
4	Try the door handle to confirm it is locked.

✓You can lock and unlock your locker as many times as needed during your session.

Auto-Lock Warning: If your locker is left OPEN (unlocked) for more than 5 minutes, the system will automatically lock it and release it back to the available pool. Make sure to keep the door closed and locked when not actively using it.

5.3 Checking the LCD Screen

The 16×2 LCD screen on the locker cabinet shows the current status of each locker bay in real time. Common messages include:

- "Available" — the locker is free for anyone to claim
- "Assigned to [User]" — the locker has been claimed but not yet unlocked
- "Locker [N] Unlocked" — the door is currently open
- "Locker [N] Locked" — the door is secured

6. Returning (Releasing) Your Locker

When you are done with the locker, you must release it so the next person can use it.

Releasing is a two-part process: retrieve your belongings, then formally return the locker.

6.1 Step-by-Step — Returning

Step	Action
1	Go back to the website and verify your identity again (so the system confirms it is you).
2	The system re-assigns you to your locker (e.g. "You have been re-assigned to Locker 1").
3	Press "Unlock" to open your locker one final time.
4	Retrieve all your belongings from the locker.
5	Close the locker door.
6	Press "I'm Done / Return Locker" on the web interface.
7	The locker status changes to "Available" on the LCD screen — the locker is now free for the next user.

Reminder: Always retrieve your belongings before pressing "Return Locker". Once released, another user may be assigned your locker immediately.

After a successful release, the LCD screen will show "Available" for that locker bay.

7. Session Restore (Page Refresh)

If you accidentally close or refresh the website while a locker is assigned to you, your session is automatically restored.

7.1 How It Works

The system saves your session in your browser's local storage. When you revisit the website, it checks the server to confirm your locker is still assigned and resumes from where you left off — no need to re-claim.

7.2 What To Do If Your Session Is Not Restored

- Close and re-open the browser tab.
- Navigate back to the Smart Locker website.
- Verify your identity when prompted.
- If the locker is still within its lease window, you will be taken back to your assigned locker automatically.

If Your Session Expired: If more than 5 minutes passed while the locker was in "Assigned" state (not yet unlocked), the locker was automatically released. Simply claim a new locker from the beginning.

8. Locker Status Lifecycle

Every locker moves through the following states:

Status	Trigger	Meaning
AVAILABLE	Initial / released	Locker is free — anyone can claim it
ASSIGNED	User claims locker	Locker is reserved but not yet physically opened
OCCUPIED	User unlocks	Locker is in active use
AVAILABLE	User releases	Locker is returned to the free pool

8.1 Automatic Timeouts

The system has two automatic safeguards to prevent lockers from getting stuck:

- **ASSIGNED → AVAILABLE:** If a claimed locker is not unlocked within 5 minutes, it automatically reverts to Available.
- **OCCUPIED → AVAILABLE:** If an open locker is left unlocked for more than 5 minutes, it is automatically locked and released back to the pool.

9. Troubleshooting

9.1 Common Issues and Solutions

The website is not loading

- Check your internet connection.
- Try refreshing the page (Ctrl + R or pull-to-refresh on mobile).
- The server may be waking up — it can take up to 30 seconds on first load (Render.com free tier cold start).

Biometric prompt is not appearing

- Ensure your browser supports WebAuthn (use Chrome, Edge, or Safari — latest version).
- Check that your device has a biometric sensor enabled and set up in your device settings.
- Try a different browser.

Locker was not assigned after claiming

- All lockers may currently be occupied. Wait a moment and try again — the auto-timeout will free any abandoned lockers.
- Refresh the page and try claiming again.

Locker bolt is not moving when I click Unlock / Lock

- Check that the ESP32 and relay are powered on (LEDs on the relay board should be visible).

- Ensure the locker cabinet is connected to power (battery pack or mains).
- Check that the ESP32 is connected to Wi-Fi — the LCD may display a connection status.
- Contact a system administrator if the hardware appears unresponsive.

I was assigned a locker but it expired (5-minute timeout)

- Simply go back to the website and claim a new locker from the beginning.
- No penalty applies — the system frees and re-assigns automatically.

I left my belongings in the locker and the locker auto-released

- Go to the website and verify your identity.
- Claim the locker again — if your belongings are still inside, you will need a system administrator to manually release the locker.
- Contact library/facility staff for assistance.

Contact: For hardware faults or access emergencies, contact your library or facility administrator. The physical locker can be manually overridden by authorised staff.

10. Quick Reference Card

Cut or photograph this section for quick access at the locker.

Task	Action
First-time setup	Go to website → Register → Complete biometric/PIN prompt
Get a locker	Website → Verify identity → Claim Locker
Open your locker	Website → Unlock → hear click → door opens
Close your locker	Website → Lock → door secured
Return locker	Website → Verify → Unlock → take items → Return Locker
Session restore	Revisit website → verify → session auto-resumes
Auto-timeout	5 min unclaimed or 5 min left open → auto-released

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Operation Video:

<https://drive.google.com/file/d/1R2BMD9DogCVqviRVV4EwPOxApFnTIgfG/view?usp=sharing>